

Simple Ticket Alert — MVP

The **smallest working version**. Goal: when a watched World Cup match has tickets available, send everyone a **browser push notification** so they can go buy them **themselves, manually**. No accounts, no payments, no bot.

This is the legal, fair version: **we alert humans — we never auto-buy**.

1. What it does (in one picture)

```
Cron (every few minutes)
  | triggers
  v
Checker → looks at the official ticket page for each watched match
          | "Is there a Buy / Available status?"
          v
Available? → send Web Push to everyone subscribed → "🎟 Tickets available: Match X!"
```

Almost identical to the goal-alert app already in this folder — we just swap "goal scored" for "tickets available".

2. What we need

Piece	Choice for the MVP	Why
Framework	Next.js (already set up here)	Reuse what we built
Notifications	Web Push (already working)	Instant, free, no email service needed
Storage	Upstash Redis (free tier)	Tiny: just store subscriptions + last status
Scheduler	cron-job.org (free) pings <code>/api/check-tickets</code>	Works even on free hosting
Data source	Ticketmaster Discovery API (+ SeatGeek for resale)	Real API, free tier — see section 4
Hosting	Vercel (free)	Free HTTPS (Web Push needs HTTPS)

Cost: \$0 to start.

3. The data we store (very small)

```
subscriptions : [ {endpoint, keys...} ] // who to notify (browsers)
matches       : [ {id, name, url} ]     // which matches we watch
lastStatus    : { matchId: "AVAILABLE" | "SOLD_OUT" } // to avoid repeat alerts
```

We only notify when status **changes** from sold-out → available (so people aren't spammed every check).

4. Where the "tickets available" info comes from

There is **no official FIFA tickets API**, but we don't need to scrape — there are **real ticketing APIs** we can use:

API	Covers	Free tier	Auth
Ticketmaster Discovery API	Official first-party tickets (Ticketmaster is FIFA's US venue partner — most of the 78 US matches)	~5,000 requests/day	API key
SeatGeek Platform API	Resale / secondary market (StubHub, Vivid Seats...) + fair-price score	Yes	API key

For the MVP, **Ticketmaster Discovery API** alone is enough.

- Endpoint pattern: `GET https://app.ticketmaster.com/discovery/v2/events.json?keyword=World+Cup&apikey=YOUR_KEY`
- Get a free key at <https://developer.ticketmaster.com/>
- We read the event's availability/status and the official `url` to buy.

Rules to stay legal & safe:

-  We only **read availability** through the API and **alert** humans.
-  The alert links to the **official purchase page** — the human buys it.
-  Check at a sensible rate (every few minutes) to stay under the free limit.
-  Never auto-login, auto-buy, or create multiple accounts. Ticketmaster's terms forbid bot buying too — that's the illegal part we never touch.

5. Build steps (what to code)

1. `/api/subscribe` — already exists. Saves a browser subscription. ✓
2. `lib/tickets.js` — `getTicketStatus(match)` : call the **Ticketmaster Discovery API** for that match and return `"AVAILABLE"` or `"SOLD_OUT"` (+ the official buy `url`).
3. `/api/check-tickets` — for each watched match: get status; if it changed to `AVAILABLE`, push an alert (with the buy link) to all subscriptions; save new status.
4. **Frontend** — list the matches being watched + the "Enable alerts" button (reuse the existing button component).
5. **Cron** — cron-job.org calls `/api/check-tickets` every 3–5 minutes.
6. **Env var** — add `TICKETMASTER_API_KEY` (like we did with the football key).

6. Notification example

 **Tickets available!** Final — MetLife Stadium. Tap to open the official site and buy now.

Tapping the notification opens the **official ticket page** so the user buys it themselves.

7. What this MVP does NOT have (on purpose)

- ❌ User accounts / login (everyone gets the same match alerts)
- ❌ Choosing specific matches per user
- ❌ Email / WhatsApp (push only)
- ❌ Payments

All of that lives in the bigger plan → see **PROJECT.md**.

8. Definition of done

- ☐ I can open the site and click "Enable alerts"
- ☐ The cron hits `/api/check-tickets` automatically
- ☐ When a watched match flips to AVAILABLE, I get a push within ~3 min
- ☐ I don't get spammed when nothing changes